

Frontend–Blockchain State Desynchronization After Successful NFT Mint

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Date	Sunday, 10 May 2026
Environment	WSL Ubuntu Node (RPC :8546) + Rabby Wallet — Indonesia / CGNAT
Report Type	Application Layer Consistency Investigation — Indexer / Frontend Synchronization
Relates to	EIP-1559 Compatibility & Txpool Analysis (09 May 2026)

Executive Summary

This report documents an application-layer state desynchronization issue observed on DAC Testnet following a successful NFT badge mint transaction. Although the mint transaction was finalized on-chain and confirmed through the DAC blockchain explorer — with the NFT verifiably present in the wallet — the frontend dashboard continued to display the badge as unminted, with the mint button remaining active.

The findings indicate a synchronization delay or failure between blockchain state, backend indexer infrastructure, and frontend UI rendering. This creates a misleading user experience where on-chain reality is not reflected in the application interface.

Layer	Status	Detail
Blockchain / On-chain layer	Operational	TX finalized, NFT confirmed
DAC Blockchain Explorer	Accurate	NFT ownership visible
Frontend dashboard (badge status)	Desynchronized	Still showing unminted
Backend indexer / event listener	Delayed / Failed	Mint event not processed

Environment

Node	WSL Ubuntu — local dacnode binary
RPC Endpoint	127.0.0.1:8546 (local node) · rpctest.dachain.tech (official)
Wallet	Rabby Wallet
Wallet Address	0x870ad63acc507cdfd878F170606d19ae78988AFE
NFT Contract	0xB36ab4c2Bd6aCfC36e9D6c53F39F4301901Bd647
Target	DAC Inception Badge System
Network	Indonesia — Residential CGNAT

Observed Behavior

After submitting the NFT badge mint transaction, the following state was observed across different layers of the DAC infrastructure:

Timeline

Date	Event	Detail
May 4, 2026 — 21:18:32	NFT Minted	TX 0x91a41b... confirmed on-chain — DAC Inception Rank token minted to wallet
May 4–10, 2026	UI Stale (6 days)	Frontend dashboard continues displaying badge as unminted — mint button remains active
May 10, 2026 — 22:12:02	Duplicate Attempt	TX 0x383c84... submitted — contract rejected (NFT already minted) → Status: Failed

Layer Status at Time of Investigation

Component	Status	Notes
Original mint TX (May 4)	Confirmed	0x91a41b... — Success
NFT ownership on-chain	Confirmed	5 NFTs present in wallet
DAC blockchain explorer	Accurate	NFT ownership visible
Frontend badge status	Stale — 6 days	Still showing unminted since May 4
Mint button UI state	Incorrect	Active despite completed mint
Duplicate mint attempt (May 10)	Failed	0x383c84... — contract rejected duplicate
UI synchronization	Failed	No update across 6-day window

Transaction Evidence

TX 1 — Original Successful Mint (May 4, 2026)

The original NFT badge mint transaction was submitted and finalized on May 4, 2026 — 6 days prior to the investigation date. The explorer confirms successful execution with full gas consumption and token transfer to the wallet.

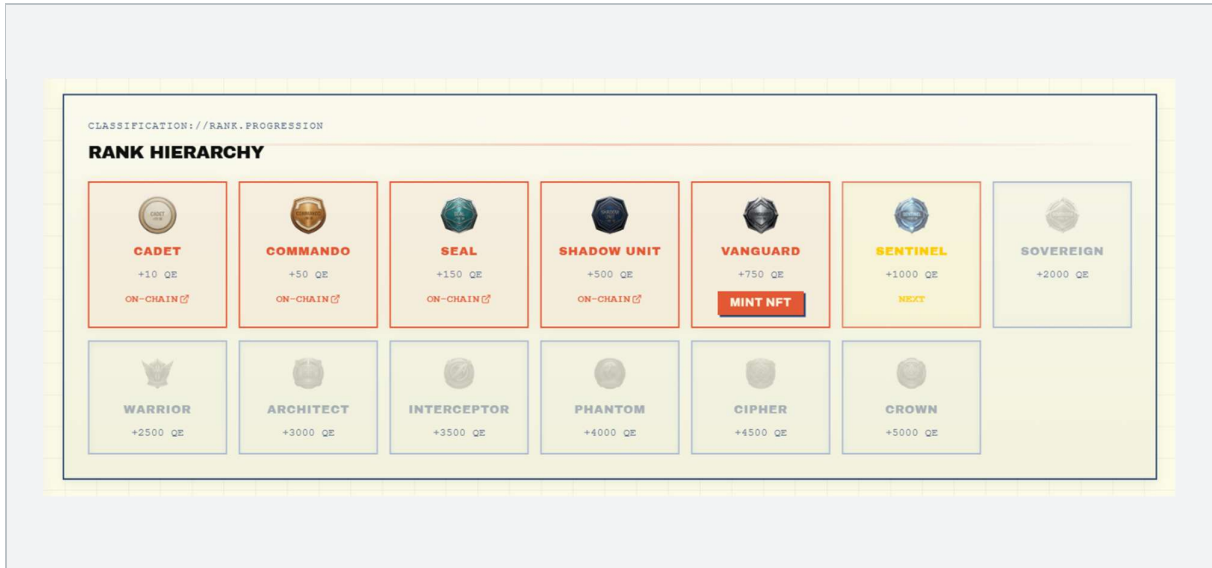
TX Hash	0x91a41be3deec3b65b0c975afe841a56566abef28e47f29bc43efea9e32f7b3aa
Status	Success

Supporting Evidence — Screenshots

The following screenshots provide visual proof of the state desynchronization. The contrast between explorer state and frontend state is the core evidence of this report.

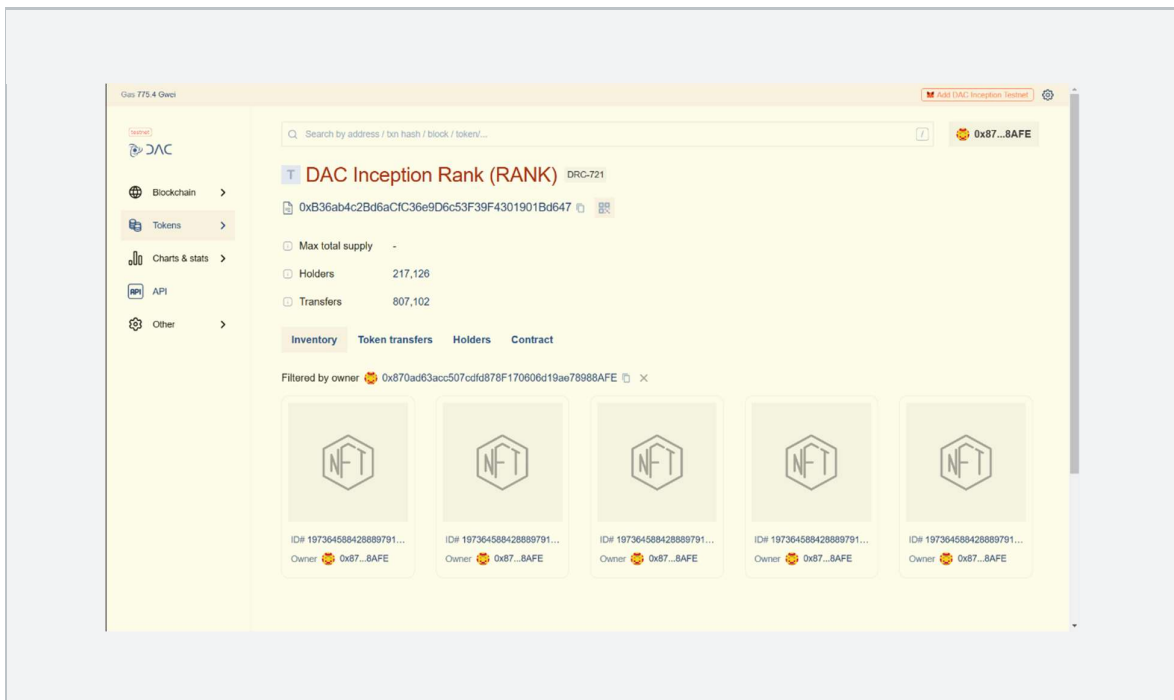
Screenshot 1 — Web Dashboard: Badge Hierarchy State

Shows the DAC Inception badge hierarchy. Other badges correctly display ON-CHAIN status with explorer links. One badge remains incorrectly marked as ready to mint despite being confirmed on-chain.



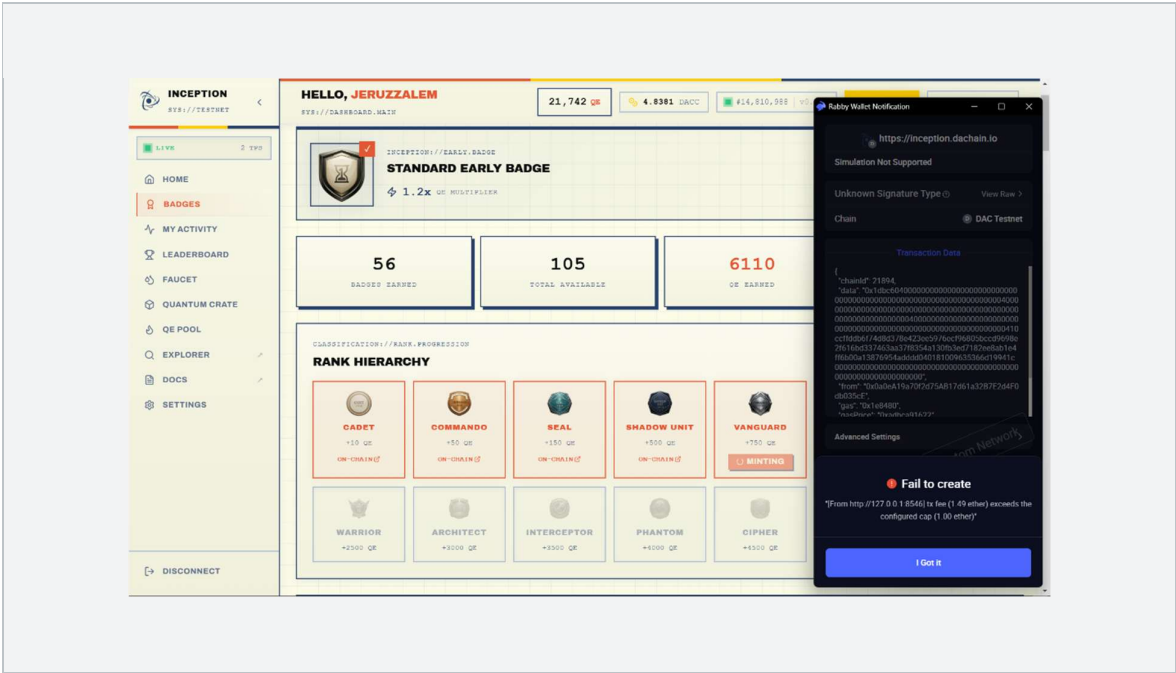
Screenshot 2 — Explorer: NFT Confirmed On-Chain

DAC blockchain explorer view of address 0x870ad6...AFE showing NFT ownership confirmed. 5 NFTs present in wallet — verifying successful on-chain mint.



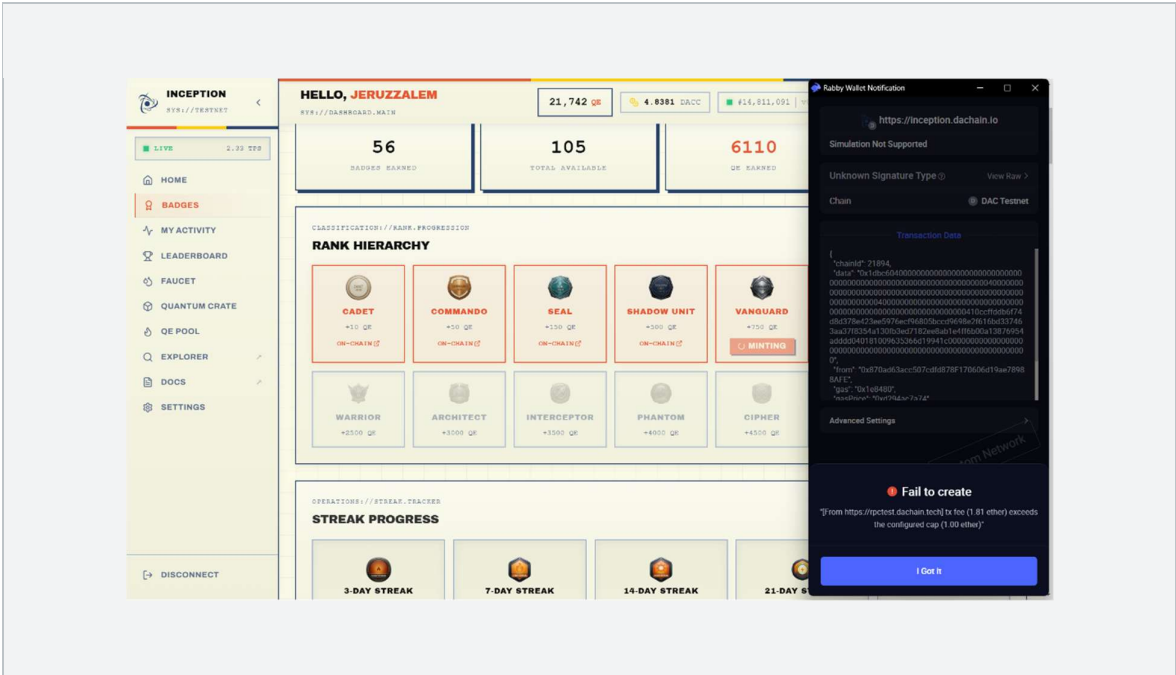
Screenshot 3 — Mint Process via Local RPC (127.0.0.1:8546)

Mint transaction flow using self-hosted local RPC. Shows Rabby Wallet interaction and transaction submission through the WSL node endpoint.



Screenshot 4 — Mint Process via Official RPC (rpctest.dachain.tech)

Mint transaction flow using the official public RPC endpoint as comparison. Confirms the desynchronization issue is not RPC-dependent — frontend state remains stale regardless of RPC source.



Technical Analysis — Possible Root Causes

The blockchain layer operated correctly and finalized the mint transaction. The desynchronization exists at the application layer. The following root causes are identified in order of likelihood:

- **Indexer Delay** — The frontend depends on a backend indexer rather than querying contract state directly. If the indexer has not processed the mint event, the UI reflects stale pre-mint state regardless of on-chain reality.
- **Event Listener Failure** — The mint event emitted by the NFT contract may not have been captured or processed correctly by the backend event listener, leaving the indexer database unchanged.
- **Cached Session State** — The frontend wallet session or browser cache may retain stale badge status from before the mint, not re-querying the backend on page load.
- **RPC Read Inconsistency** — The RPC endpoint serving the frontend may be lagging behind the canonical chain state, returning outdated contract reads for badge ownership checks.

Note on `eth_estimateGas` Revert

The `eth_estimateGas` execution revert observed in the node log is a separate and secondary issue. The custom error selector `0xf3998cd1` returned in `errdata` is a Solidity custom error — likely a precondition check within the NFT contract that fires during simulation (e.g. ownership guard, duplicate mint check, or signature validation). This does not indicate transaction failure. The transaction was submitted and finalized successfully despite simulation failure.

User Impact

This desynchronization issue represents a significant user experience failure that is likely not isolated to a single user or incident. The UI bug may have persisted for an extended and indeterminate period — making it highly probable that a large portion of the community is currently affected: NFT already minted and confirmed on-chain, yet the frontend continuously displays the badge as unminted. Community members began raising awareness of this issue on Sunday, 10 May 2026, noting that many successful mints were going unrecognized due to the stale UI. The following impacts have been identified:

- Users who minted successfully are unaware of their NFT ownership due to stale UI — the dashboard does not reflect on-chain reality.
- The active mint button on an already-minted badge directly causes failed duplicate transactions, resulting in unnecessary gas fee losses on each attempt.
- Users may have submitted multiple failed mint attempts over an extended and unknown period — each consuming gas fees with no benefit, without realizing the original mint had already succeeded.
- Community members experiencing this issue may interpret the persistent mint button as a sign of a failed mint, not a UI bug — leading to continued retry behavior.
- Badge hierarchy progression and rank display are visually inconsistent with actual on-chain holdings, affecting trust in the system.
- Without a mechanism to detect or surface this desynchronization, affected users have no self-service path to resolution.

Recommendations

For DAC Team — Application Layer

- Implement direct on-chain contract state verification as fallback after mint completion, independent of indexer state.

- Add transaction receipt validation before frontend state rendering — UI should update on TX confirmation, not indexer confirmation.
- Display a pending synchronization indicator (e.g. "Confirming on-chain...") instead of retaining the stale mint button.
- Add a manual refresh or re-sync mechanism in the dashboard UI for users to trigger a state update.
- Improve indexer monitoring and event synchronization observability — alert on processing lag.

For Testnet Participants

- Verify mint success via the DAC blockchain explorer before attempting any follow-up mint action.
- Perform a hard refresh (Ctrl + Shift + R) and disconnect/reconnect wallet if UI appears stale.
- Do not re-attempt minting if the explorer confirms the NFT is already present in your wallet.

Conclusion

Layer	Status
Blockchain / on-chain layer	Operational — TX finalized, NFT confirmed
Smart contract execution	Successful — mint completed despite estimateGas simulation failure
Backend indexer / event sync	Delayed or incomplete — mint event not reflected in frontend
Frontend UI state	Desynchronized — stale state caused unnecessary failed duplicate transactions
Community impact	Likely widespread — multiple users may be affected by same desync

The DAC blockchain layer operated correctly throughout this investigation. The original mint transaction was finalized on-chain with full confirmation. The inconsistency exists entirely at the application layer — the frontend dashboard failed to synchronize with verified on-chain state, during which the stale UI actively misled users into submitting duplicate mint transactions that failed and consumed gas fees. Given the community-wide nature of this badge system, it is highly probable that a significant number of users are currently affected by the same desynchronization, unknowingly retrying already-completed mints.